

0620 • PAPER 4

TOPIC

02

Atoms, elements and compounds

Questions only • Original examination pages retained

59

QUESTIONS

711

TOTAL MARKS

1

SECTIONS



USE THE TRACKER • ATTEMPT WITHOUT NOTES • MARK • REVIEW ERRORS

TOPIC CONTENTS

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01

Atomic structure, isotopes, ions, bonding and structures

59 questions • 711 marks

2.1 Atomic structure, isotopes, ions, bonding and structures

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2.1 Atomic structure, isotopes, ions, bonding and structures

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2.1 Atomic structure, isotopes, ions, bonding and structures

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2.1 Atomic structure, isotopes, ions, bonding and structures

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2.1

Atomic structure, isotopes, ions, bonding and structures

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3 The Periodic Table is a method of classifying elements.

(a) Identify the element which is in Group VI and Period 4.

..... [1]

(b) Calcium is in Group II and chlorine is in Group VII of the Periodic Table.

Explain, in terms of number of outer shell electrons and electron transfer, how calcium atoms and chlorine atoms form ions. Give the formulae of the ions formed.

.....
.....
.....
.....
.....
.....
..... [5]

(c) Group V chlorides are covalent molecules. The boiling points of some Group V chlorides are shown.

chloride	boiling point/°C
NCl_3	71
PCl_3	
AsCl_3	130
SbCl_3	283

(i) Suggest the approximate boiling point of PCl_3 .

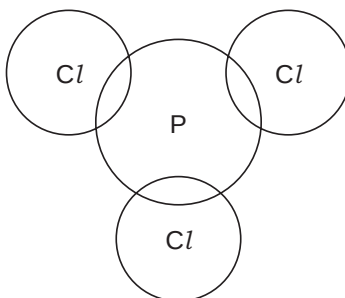
..... [1]

(ii) Explain the trend in boiling points in terms of attractive forces between particles.

.....
..... [2]

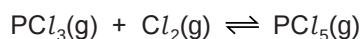
- (iii) Complete the dot-and-cross diagram to show the electron arrangement in a molecule of PCl_3 .

Show outer electrons only.



[3]

- (d) PCl_3 reacts with chlorine, Cl_2 , to form PCl_5 . This reaction is exothermic and reaches an equilibrium.



- (i) Describe **two** features of an equilibrium.

.....

 [2]

- (ii) State the effect, if any, on the position of this equilibrium when the following changes are made.
 Explain your answers.

temperature is increased

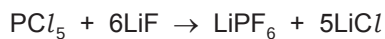
 pressure is increased
 [4]

- (iii) Explain, in terms of particles, what happens to the rate of the forward reaction when the reaction mixture is heated.

.....

 [3]

- (e) PCl_5 reacts with lithium fluoride, LiF , to form LiPF_6 .



Calculate the mass of LiF needed to form 3.04 g of LiPF_6 using the following steps.

- Calculate the number of moles of LiPF_6 formed.
[M_r : LiPF_6 , 152]

number of moles =

- Deduce the number of moles of LiF needed.

number of moles =

- Calculate the mass of LiF needed.

mass = g
[3]

- (f) Lithium fluoride has ionic bonding.

- (i) What is an ionic bond?

.....
..... [2]

- (ii) Give **two** physical properties of ionic compounds.

.....
..... [2]

[Total: 28]

1 This question is about elements X, Y and Z.

(a) An atom of element X is represented as ${}_{16}^{34}\text{X}$.

(i) Name the different types of particles found in the nucleus of this atom of X.

.....
..... [2]

(ii) What is the term for the total number of particles in the nucleus of an atom?

..... [1]

(iii) What is the total number of particles in the nucleus of an atom of ${}_{16}^{34}\text{X}$?

..... [1]

(iv) What is the electronic structure of the ion X^{2-} ?

..... [1]

(v) Suggest the formula of the compound formed between aluminium and X.

..... [1]

(b) (i) What term is used to describe atoms of the same element with different numbers of particles in the nucleus?

..... [1]

(ii) Identify the atom against which the relative masses of all other atoms are compared.

..... [1]

(iii) What is the name of the amount of any substance that contains 6.02×10^{23} particles?

..... [1]

(iv) The constant 6.02×10^{23} has a name.

What is the name of this constant?

..... [1]

- (c) Part of the definition of relative atomic mass is 'the average mass of naturally occurring atoms of an element'.

Some relative atomic masses are not whole numbers.

Element Y has only two different types of atom, ^{69}Y and ^{71}Y .

The ratio of atoms present in element Y is shown.

$$^{69}\text{Y} : ^{71}\text{Y} = 3 : 2$$

- Calculate the relative atomic mass of element Y to **one decimal place**.

relative atomic mass =

- Identify element Y.

.....
[3]

- (d) Element Z is in Period 3 and Group V.

- (i) Identify element Z.

..... [1]

- (ii) Explain in terms of electron transfer why Z behaves chemically as a non-metal.

.....
..... [2]

[Total: 16]

2 Magnesium is a metal.

(a) Name and describe the bonding in magnesium.

name

description of bonding

.....

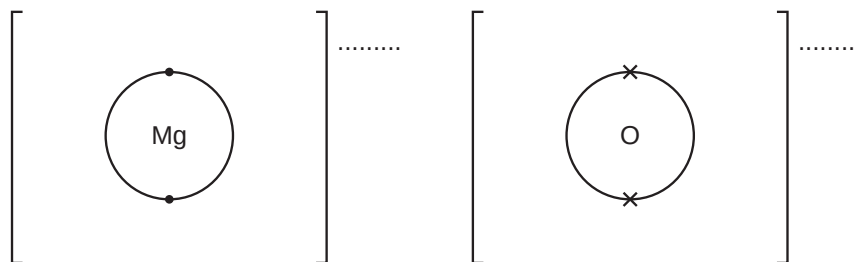
.....

[4]

(b) Magnesium oxide, MgO , is formed when magnesium burns in oxygen.

(i) Complete the dot-and-cross diagram to show the electron arrangement of the ions in magnesium oxide.

The inner shells have been drawn.
Give the charges on the ions.



[3]

(ii) Write the chemical equation for the reaction that occurs when magnesium burns in oxygen.

..... [2]

(c) Magnesium oxide also forms when magnesium nitrate, $\text{Mg}(\text{NO}_3)_2$, is heated strongly. This is an endothermic reaction.

(i) Write the chemical equation for this reaction.

..... [2]

(ii) What type of reaction is this?

..... [1]

(iii) Name **two** other compounds of magnesium that form magnesium oxide when heated.

.....

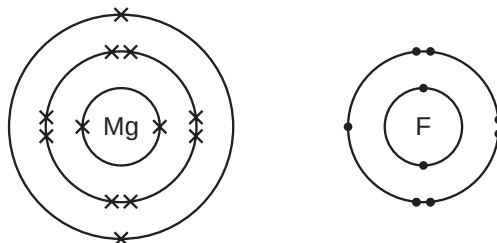
..... [2]

[Total: 14]

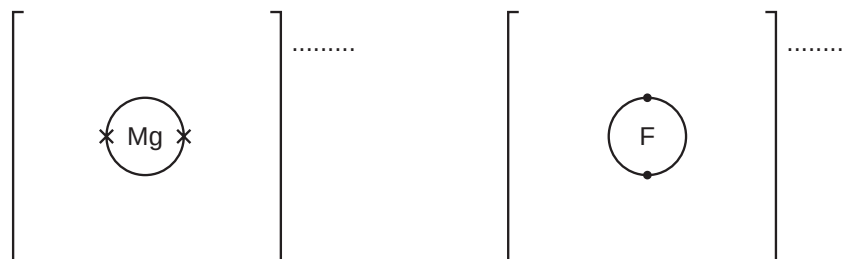
2 Fluorine forms both ionic and covalent compounds.

(a) Magnesium reacts with fluorine to form the ionic compound magnesium fluoride.

The electronic structures of an atom of magnesium and an atom of fluorine are shown.



(i) Complete the dot-and-cross diagrams to show the electronic structures of one magnesium ion and one fluoride ion. Show the charges on the ions.



[3]

(ii) What is the formula of magnesium fluoride?

..... [1]

(iii) Magnesium fluoride does **not** conduct electricity when it is solid.

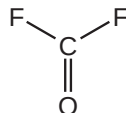
What can be done to solid magnesium fluoride to make it conduct electricity?

In your answer explain why magnesium fluoride conducts electricity when this change is made.

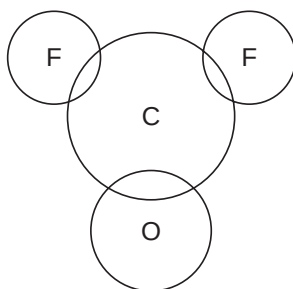
.....

 [2]

- (b) Carbonyl fluoride, COF_2 , is a covalent compound. The structure of a molecule of COF_2 is shown.



Complete the dot-and-cross diagram to show the electron arrangement in a molecule of carbonyl fluoride. Show outer shell electrons only.



[3]

- (c) The melting points of magnesium fluoride and carbonyl fluoride are shown.

	melting point/ $^{\circ}\text{C}$
magnesium fluoride	1263
carbonyl fluoride	-111

- (i) Explain, using your knowledge of structure and bonding, why magnesium fluoride has a high melting point.

.....

 [2]

- (ii) Explain, using your knowledge of structure and bonding, why carbonyl fluoride has a low melting point.

.....

 [2]

[Total: 13]

3 Chlorine is in Group VII of the Periodic Table.

(a) Two isotopes of chlorine are chlorine-35 and chlorine-37.

(i) State why these two isotopes of chlorine have the same chemical properties.

.....

 [2]

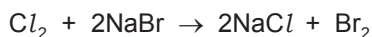
(ii) Complete the table to show the number of electrons, neutrons and protons in each atom and ion.

	number of electrons	number of neutrons	number of protons
$^{35}_{17}\text{Cl}$			
$^{37}_{17}\text{Cl}^-$			

[3]

(b) (i) Chlorine reacts with aqueous sodium bromide.

The equation for the reaction is shown.



State the type of reaction shown.

..... [1]

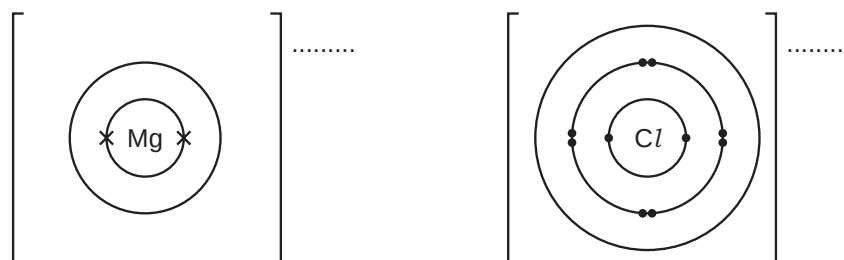
(ii) Why is there **no** reaction between iodine and aqueous sodium bromide?

..... [1]

(c) Magnesium reacts with chlorine to form magnesium chloride.

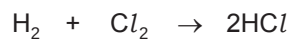
Complete the dot-and-cross diagram to show the electron arrangement of the ions in magnesium chloride. Give the charges on the ions.

The inner shells have been completed.

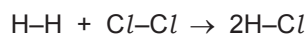


[3]

(d) Hydrogen and chlorine react to form hydrogen chloride gas, as shown in the equation.



This equation can be represented as shown.



Some bond energies are shown in the table.

bond	bond energy in kJ/mol
H-H	436
Cl-Cl	243
H-Cl	432

Calculate the energy change for the reaction between hydrogen and chlorine, using the following steps.

- Calculate the energy needed to break the bonds.

..... kJ

- Calculate the energy released when bonds are formed.

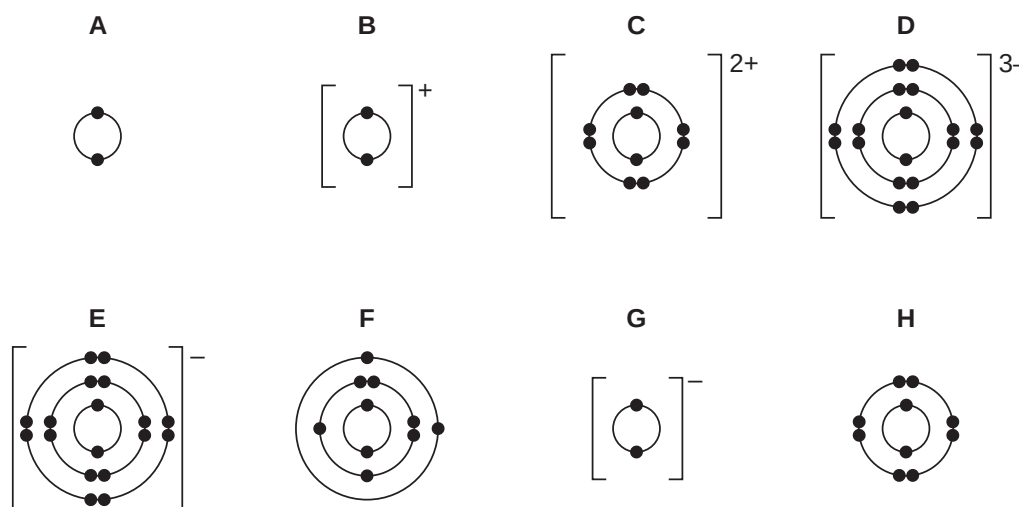
..... kJ

- Calculate the energy change for the reaction.

..... kJ/mol
[3]

[Total: 13]

1 The electronic structures of some atoms and ions are shown.



(a) Write the letters, **A, B, C, D, E, F, G** or **H**, of the electronic structures which show:

- (i) atoms of two different noble gases and [2]
- (ii) an ion of a Group I element [1]
- (iii) an ion of a Group V element [1]
- (iv) a pair of ions that could form a compound with the formula XY_2 , and [1]

(b) State which electronic structure, **A, B, C, D, E, F, G** or **H**, is incorrect.

Explain why.

incorrect electronic structure

explanation

[2]

(c) State how many protons are found in the nucleus of ion **C**. [1]

(d) Use the Periodic Table to deduce:

- (i) the chemical symbol for ion **G** [1]
- (ii) the element which forms an ion with a 3+ charge and the same electronic structure as **H**.
..... [1]

[Total: 10]

- 2 The table gives information about five particles, **A**, **B**, **C**, **D** and **E**.

particle	number of electrons	number of neutrons	number of protons
A	10	13	11
B	18	20	18
C	18	18	18
D	10	12	8
E	10	10	10

- (a) State the atomic number of **A**.

..... [1]

- (b) State the nucleon number of **B**.

..... [1]

- (c) Write the electronic structure of **C**.

..... [1]

- (d) Give the letters of all the particles which are:

(i) atoms [1]

(ii) positive ions [1]

(iii) negative ions [1]

(iv) isotopes of each other. [1]

[Total: 7]

- 1 The table shows the numbers of protons, neutrons and electrons in particles **A** to **I**.

particle	protons	neutrons	electrons
A	1	0	0
B	6	6	6
C	6	8	6
D	10	10	10
E	16	16	18
F	17	18	17
G	18	22	18
H	19	20	19
I	20	20	18

Answer the following questions about particles **A** to **I**. Each letter may be used once, more than once or not at all.

- (a) State which of the particles **A** to **I**:

- (i) is an anion [1]
- (ii) are cations and [2]
- (iii) are noble gas atoms and [2]
- (iv) is a halogen atom [1]
- (v) is a Group I atom [1]
- (vi) have the same nucleon number and [1]
- (vii) causes acidity in aqueous solutions [1]
- (viii) is used to define the relative atomic mass of elements. [1]

- (b) Explain why **B** and **C** are isotopes of the same element.

.....

..... [2]

[Total: 12]

2 Complete the table to:

- deduce the number of protons, electrons and neutrons in the magnesium atom and copper ion shown
- identify the atom or ion represented by the final row.

	number of protons	number of electrons	number of neutrons
$^{25}_{12}\text{Mg}$	12		
$^{65}_{29}\text{Cu}^{2+}$			36
	17	18	20

[Total: 5]

- 1 The symbols of the elements of Period 3 of the Periodic Table are shown.

Na	Mg	Al	Si	P	S	Cl	Ar
----	----	----	----	---	---	----	----

Answer the following questions about these elements.

Each element may be used once, more than once or not at all.

Write the symbol of an element which:

- (a) is malleable

..... [1]

- (b) has only two electrons in its outermost shell

..... [1]

- (c) forms an oxide which leads to acid rain

..... [1]

- (d) forms an ion with a 2– charge

..... [1]

- (e) is extracted from an ore called bauxite

..... [1]

- (f) does **not** form an oxide

..... [1]

- (g) forms an oxide with a macromolecular structure

..... [1]

- (h) forms an amphoteric oxide

..... [1]

- (i) exists as diatomic molecules

..... [1]

- (j) forms a binary compound with hydrogen that is a strong acid.

..... [1]

[Total: 10]

2 Silver has an atomic number of 47.

(a) Naturally occurring atoms of silver are $^{107}_{47}\text{Ag}$ and $^{109}_{47}\text{Ag}$.

(i) State the name given to atoms of the same element with different nucleon numbers.

..... [1]

(ii) Complete the table to show the number of protons, neutrons and electrons in each atom and ion of silver shown.

	$^{107}_{47}\text{Ag}$	$^{109}_{47}\text{Ag}^+$
protons		
neutrons		
electrons		

[3]

(iii) Complete this definition of relative atomic mass.

Relative atomic mass is the mass of naturally occurring atoms of an element on a scale where the atom has a mass of exactly units.

[3]

(iv) A sample of silver has a relative atomic mass of 108.0.

Deduce the percentage of ^{107}Ag present in this sample of silver.

..... [1]

(b) Silver nitrate is a salt of silver made by reacting silver oxide with an acid.

Write the formula of the acid which reacts with silver oxide to form silver nitrate.

..... [1]

(c) Aqueous silver nitrate is a colourless solution containing $\text{Ag}^+(\text{aq})$ ions.

(i) Describe what is seen when aqueous silver nitrate is added to aqueous sodium iodide, $\text{NaI}(\text{aq})$.

..... [1]

(ii) Write the ionic equation for the reaction between aqueous silver nitrate and aqueous sodium iodide.
Include state symbols.

..... [3]

(d) In the positive test for aqueous nitrate ions, aqueous sodium hydroxide and one other substance are warmed with the nitrate ions.

Name this other substance and the gas formed.

name of substance

name of gas [2]

(e) When silver nitrate is exposed to sunlight, silver is formed.

Name the type of reaction which needs light to make it happen.

..... [1]

(f) Members of one homologous series only react with chlorine in the presence of sunlight.

(i) Name a member of this homologous series.

..... [1]

(ii) Name **two** products that form when the compound in (i) reacts with chlorine.

1

2 [2]

[Total: 19]

5 This question is about compounds of nitrogen.

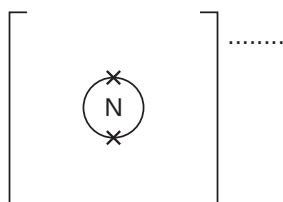
(a) Nitrogen reacts with lithium to form lithium nitride, Li_3N .

(i) Write the chemical equation for the reaction between lithium and nitrogen.

..... [2]

(ii) Lithium nitride is ionically bonded.

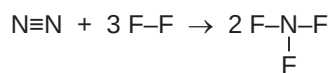
Complete the diagram to show the electronic structure of the nitride ion.
Show the charge on the nitride ion.



[2]

(b) Nitrogen reacts with fluorine to form nitrogen trifluoride, NF_3 .

(i) The chemical equation can be represented as shown.



Some bond energies are shown in the table.

bond	bond energy in kJ/mol
$\text{N}\equiv\text{N}$	945
$\text{F}-\text{F}$	160
$\text{N}-\text{F}$	300

Calculate the energy change for the reaction between nitrogen and fluorine, using the following steps:

- energy taken in to break bonds

..... kJ

- energy released when bonds are formed

..... kJ

- energy change during the reaction.

..... kJ/mol
[3]

(ii) Use your answer to (i) to deduce whether this reaction is endothermic or exothermic. Explain your answer.

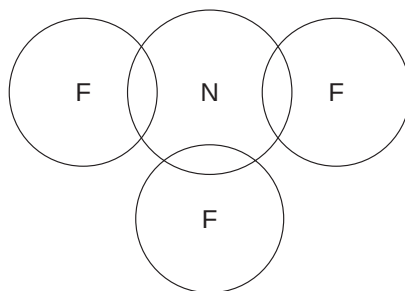
.....

..... [1]

- (iii) Complete the dot-and-cross diagram to show the electron arrangement in a molecule of NF_3 .

Use dots for nitrogen electrons and crosses for fluorine electrons.

Show outer electrons only.



[3]

- (c) Lithium nitride melts at 813°C . Nitrogen trifluoride melts at -206°C .

Explain in terms of attractive forces why lithium nitride has a much higher melting point than nitrogen trifluoride.

In your answer refer to the types of attractive forces between particles and their relative strengths.

.....
.....
.....
..... [3]

- (d) Ammonium nitrate, NH_4NO_3 , is a compound of nitrogen.

- (i) Calculate the percentage by mass of nitrogen in ammonium nitrate.

percentage by mass of nitrogen = [2]

- (ii) State a use of ammonium nitrate in agriculture.

..... [1]

- (iii) State the name of a compound that will displace ammonia from ammonium nitrate.

..... [1]

(e) Ammonia is a base which forms a weakly alkaline solution when dissolved in water.

(i) Define the term *base*.

..... [1]

(ii) Suggest the pH of aqueous ammonia.

..... [1]

[Total: 20]

2 Complete the table to:

- deduce the number of protons, electrons and neutrons in the boron atom and chloride ion shown
- identify the atom or ion represented by the final row.

formula	number of protons	number of electrons	number of neutrons
$^{11}_5\text{B}$		5	
$^{35}_{17}\text{Cl}^-$	17		
	24	21	30

[Total: 5]

2 This question is about copper and its compounds.

(a) Copper has two different naturally occurring atoms, ^{63}Cu and ^{65}Cu .

(i) State the term used for atoms of the same element with different nucleon numbers.

..... [1]

(ii) The atomic number of copper is 29.

Complete the table to show the number of protons, neutrons and electrons in the particles of copper shown.

	^{63}Cu	$^{65}\text{Cu}^{2+}$
protons		
neutrons		
electrons		

[3]

(iii) Relative atomic mass is the average mass of naturally occurring atoms of an element.

The percentage of the naturally occurring atoms in a sample of copper is shown.

^{63}Cu	^{65}Cu
70%	30%

Deduce the relative atomic mass of copper in this sample.

Give your answer to **one** decimal place.

relative atomic mass = [2]

(b) Anhydrous copper(II) sulfate is used to test for the presence of water. When this test is positive, hydrated copper(II) sulfate is formed.

(i) State the colour change seen during this test.

from to [2]

(ii) Complete the chemical equation to show the reaction that takes place.



(iii) State how hydrated copper(II) sulfate can be turned back into anhydrous copper(II) sulfate.

..... [1]

(iv) Describe a test for pure water.

.....
..... [2]

(c) Aqueous copper(II) sulfate contains $\text{Cu}^{2+}(\text{aq})$ ions.

(i) Describe what is seen when aqueous copper(II) sulfate is added to aqueous sodium hydroxide, $\text{NaOH}(\text{aq})$.

..... [1]

(ii) Write the ionic equation for the reaction between aqueous copper(II) sulfate and aqueous sodium hydroxide.

Include state symbols.

..... [3]

- (d) When solid copper(II) nitrate is heated copper(II) oxide, nitrogen dioxide and oxygen are formed.



Calculate the volume of nitrogen dioxide formed at room temperature and pressure when 4.7 g of $\text{Cu}(\text{NO}_3)_2$ is heated.

Use the following steps:

- calculate the mass of one mole of $\text{Cu}(\text{NO}_3)_2$

..... g

- calculate the number of moles of $\text{Cu}(\text{NO}_3)_2$ used

..... moles

- determine the number of moles of nitrogen dioxide formed

..... moles

- calculate the volume of nitrogen dioxide formed at room temperature and pressure.

..... dm^3
[4]

- (e) Write the chemical equation to show the action of heat on sodium nitrate, NaNO_3 .

..... [2]

[Total: 22]

3 Atoms contain protons, neutrons and electrons.

(a) Complete the table to show the relative mass and the relative charge of a proton, a neutron and an electron.

	relative mass	relative charge
proton		
neutron		
electron	$\frac{1}{1840}$	

[3]

(b) The table shows the number of protons, neutrons and electrons in some atoms and ions.

Complete the table.

atom or ion	number of protons	number of neutrons	number of electrons
$^{32}_{16}\text{S}$			
$^{39}_{19}\text{K}^+$			
	35	44	36

[5]

[Total: 8]

4 Carbon is an important element.

(a) Carbon exists as the isotopes $^{12}_6\text{C}$ and $^{13}_6\text{C}$.

Complete the table.

isotope	number of protons in one atom	number of electrons in one atom	number of neutrons in one atom
$^{12}_6\text{C}$			
$^{13}_6\text{C}$			

[2]

(b) Name **two** forms of the element carbon which have giant covalent structures.

..... and [1]

(c) The Avogadro constant is the number of particles in 1 mole.

The numerical value of the Avogadro constant is 6.02×10^{23} .

(i) Calculate the number of molecules in 22.0 g of carbon dioxide, CO_2 .

..... molecules [2]

(ii) Calculate the number of molecules in 6.00 dm^3 of carbon dioxide gas at room temperature and pressure.

..... molecules [1]

[Total: 6]

- 1 This question is about the first 30 elements in the Periodic Table.

Name the element which:

- (a) is 78% of clean, dry air [1]
- (b) has atoms with an electronic structure of 2,8,1 [1]
- (c) is extracted from hematite [1]
- (d) forms an oxide with a giant covalent structure [1]
- (e) is the gas with the slowest rate of diffusion at room temperature
..... [1]
- (f) has an anhydrous chloride which turns pink when water is added
..... [1]
- (g) has aqueous ions which form a white precipitate when added to aqueous silver ions
..... [1]
- (h) forms a blue hydroxide which dissolves in aqueous ammonia
..... [1]
- (i) is added to molten iron to remove impurities in the steel making process
..... [1]
- (j) is used to galvanise iron. [1]

[Total: 10]

- 2 A student adds excess large pieces of magnesium carbonate, MgCO_3 , to dilute hydrochloric acid, HCl , and measures the volume of carbon dioxide gas, CO_2 , given off.

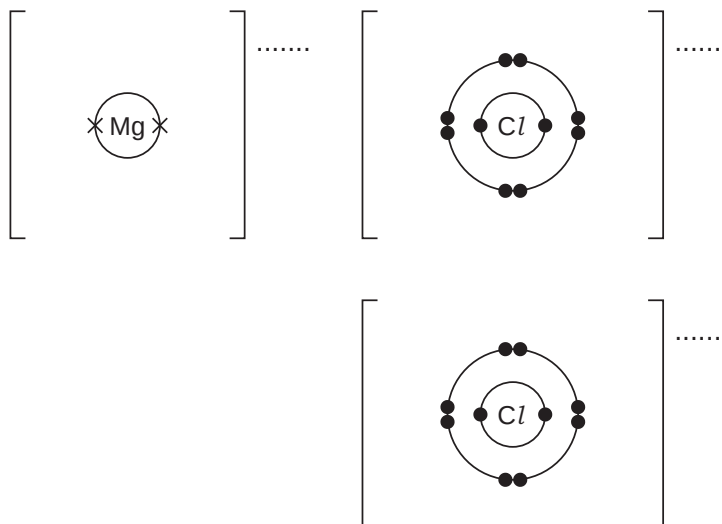
(a) Add the missing state symbols to the chemical equation for the reaction.



- (b) Complete the dot-and-cross diagram to show the electron arrangement of the ions in magnesium chloride.

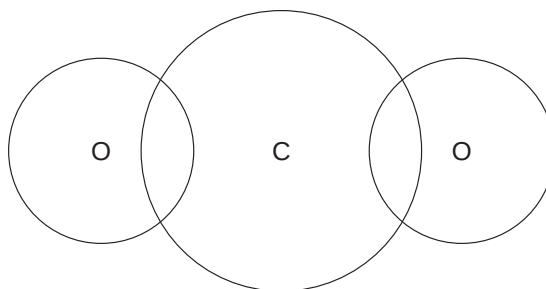
The inner shells have been drawn.

Give the charges on the ions.



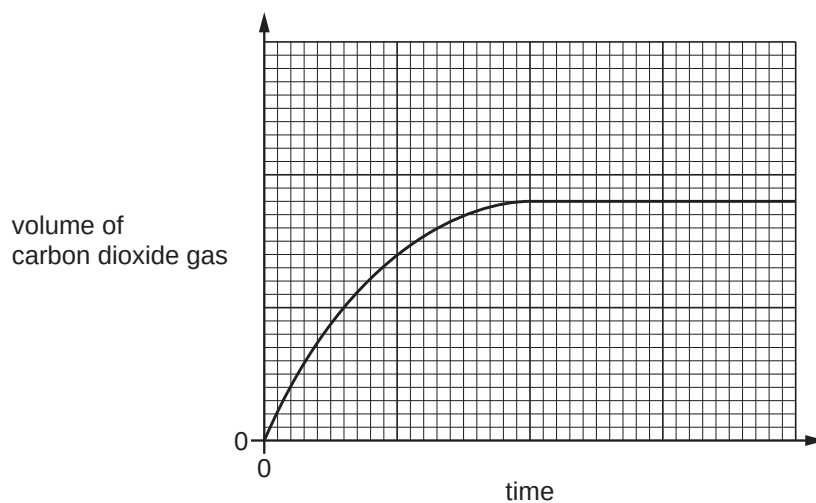
[3]

- (c) Complete the dot-and-cross diagram to show the electron arrangement in a molecule of carbon dioxide.
Show outer shell electrons only.



[2]

(d) The graph shows how the volume of carbon dioxide gas changes with time.



(i) Describe how the graph shows that the rate of this reaction decreases as time increases.

..... [1]

(ii) Explain, in terms of particles, why the rate of this reaction decreases as time increases.

..... [2]

(iii) The student repeats the experiment using powdered MgCO_3 instead of large pieces.

All other conditions stay the same.

On the grid, draw the line expected when powdered MgCO_3 is used instead of large pieces. [2]

- (e) Determine the volume of CO_2 gas given off when excess MgCO_3 is added to 25.0 cm^3 of 0.400 mol/dm^3 HCl at room temperature and pressure.



Use the following steps.

- Calculate the number of moles of HCl in 25.0 cm^3 of 0.400 mol/dm^3 of acid.

..... mol

- Determine the number of moles of CO_2 gas given off.

..... mol

- Calculate the volume of CO_2 gas given off in cm^3 .

..... cm^3
[3]

[Total: 14]

- 2 (a) Atoms are made of protons, neutrons and electrons. Atoms of the same element are known as isotopes.

(i) Complete the table.

particle	relative charge	relative mass
electron		$\frac{1}{1840}$
neutron		
proton	+1	

[2]

- (ii) $^{24}_{12}\text{Mg}$ and $^{25}_{12}\text{Mg}$ are isotopes of magnesium.

Complete the table to show the numbers of electrons, neutrons and protons in these isotopes of magnesium.

isotope	number of electrons	number of neutrons	number of protons
$^{24}_{12}\text{Mg}$			
$^{25}_{12}\text{Mg}$			

[2]

- (iii) Explain why magnesium ions have a charge of 2+.

.....
 [1]

- (b) Mg^{2+} ions have the electronic structure 2,8.

Give the formula of the following particles which have the same electronic structure as Mg^{2+} ions.

- a cation (positive ion)

.....

- an anion (negative ion)

.....

- an atom

.....

[3]

[Total: 8]

3 Transition elements are found in the middle block of the Periodic Table.

(a) Chromium has several isotopes. Manganese has only one isotope.

(i) State what is meant by the term *isotopes*.

.....
 [2]

(ii) State the nucleon number of manganese.

..... [1]

(iii) Complete the table to show the number of protons, neutrons and electrons in a $^{52}_{24}\text{Cr}^{3+}$ ion.

protons	neutrons	electrons

[3]

(b) One chemical property of transition elements is that they form coloured compounds.

(i) Give the colours of the following hydrated salts.

- hydrated copper(II) sulfate
- hydrated cobalt(II) chloride [2]

(ii) State two **other** chemical properties of transition elements.

- 1
- 2 [2]

(c) Transition elements and Group I elements are metals. They share many physical properties including the ability to:

- conduct electricity
- be hammered into shape.

(i) Explain why transition elements and Group I elements conduct electricity.

..... [1]

(ii) State the property that describes a material which can be hammered into shape.

..... [1]

- (d) Transition elements and Group I elements differ in other physical properties. Transition elements are harder and stronger than Group I elements.

Describe two **other** ways in which the physical properties of transition elements differ from Group I elements.

1

2

[2]

[Total: 14]

4 Fluorine and chlorine are halogens.

(a) Suggest the appearance of fluorine.

..... [1]

(b) Fluorine reacts with sulfur to form a compound which has 25.2% sulfur by mass and a relative molecular mass of 254.

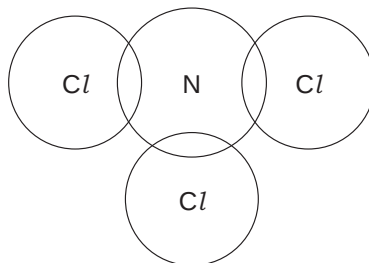
Determine the molecular formula of this compound.

molecular formula = [3]

(c) Nitrogen trichloride, NCl_3 , is a covalent compound.

Complete the dot-and-cross diagram to show the electron arrangement in a molecule of NCl_3 .

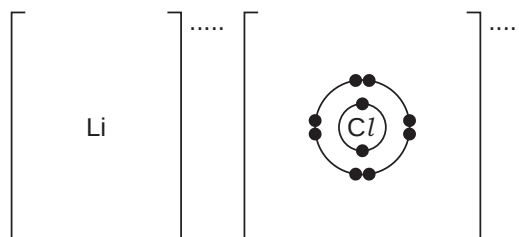
Show outer electrons only.



[3]

(d) Lithium chloride, LiCl , is an ionic compound.

Complete the dot-and-cross diagram to show the electron arrangement and charges of the ions in lithium chloride.



[3]

(e) Explain, in terms of attractive forces between particles, why LiCl is a solid at room temperature but NCl_3 is a liquid with a relatively low boiling point.

.....
.....
..... [3]

[Total: 13]

- 1 A list of substances is shown.

aluminium oxide	carbon dioxide	carbon monoxide	chlorine	copper
glucose	iron(III) oxide	limestone	nitrogen	oxygen

Answer the questions using the substances in the list.

Each substance may be used once, more than once or not at all.

State which substance is:

- (a) a product of respiration

..... [1]

- (b) the main constituent of hematite

..... [1]

- (c) an element which has a sulfate that is used to test for water

..... [1]

- (d) a colourless toxic gas

..... [1]

- (e) a reactant in fermentation

..... [1]

- (f) a reducing agent in the extraction of iron

..... [1]

- (g) a conductor of electricity when solid

..... [1]

- (h) a gas that is approximately 21% of clean, dry air.

..... [1]

[Total: 8]

- 2 (a) $^{32}_{16}\text{S}$ and $^{33}_{16}\text{S}$ are isotopes of sulfur.

Use your knowledge of protons, neutrons and electrons to answer the following questions.

- (i) Describe how these isotopes of sulfur are the same and how they are different.

same
.....
different
..... [3]

- (ii) Explain why each of these isotopes have an overall charge of zero.

.....
..... [1]

- (iii) Explain why both isotopes have the same chemical properties.

.....
..... [1]

- (b) Sulfide ions, S^{2-} , have the electronic structure 2,8,8.

- (i) Explain why sulfide ions have a charge of 2−.

.....
..... [1]

- (ii) Give the formula of:

- an anion which has the same electronic structure as S^{2-}

.....

- a cation which has the same electronic structure as S^{2-} .

..... [2]

[Total: 8]

3 This question is about nitrogen and compounds of nitrogen.

(a) Nitrogen molecules have the formula N_2 .

Some properties of nitrogen are shown:

- melting point of -210°C
- boiling point of -196°C
- non-conductor of electricity when solid
- insoluble in water.

(i) Name the type of bonding between the atoms in an N_2 molecule.

..... [1]

(ii) Explain, in terms of attractive forces between particles, why nitrogen has a low melting point.

.....
..... [1]

(iii) Explain why nitrogen does **not** conduct electricity.

.....
..... [1]

(b) Nitrogen reacts with hydrogen to form ammonia, NH_3 , in the Haber process.

State the essential conditions in the Haber process. Write an equation for the chemical reaction.

.....
.....
.....
.....
..... [4]

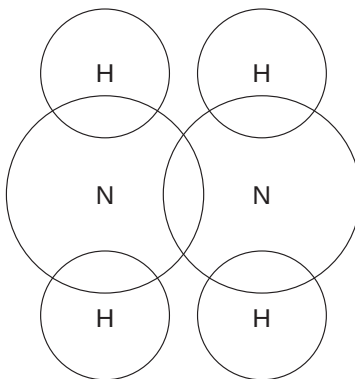
(c) Ammonia is made in the laboratory by heating ammonium chloride with calcium hydroxide.

Balance the chemical equation for the reaction.



- (d) Hydrazine, N_2H_4 , is another compound that contains nitrogen and hydrogen.

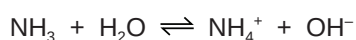
Complete the dot-and-cross diagram to show the electron arrangement in a molecule of hydrazine. Show outer electrons only.



[2]

- (e) Ammonia and hydrazine are weak bases.

The chemical equation for the reaction between one molecule of ammonia and one molecule of water is shown.



- (i) State the meaning of the term *base*.

..... [1]

- (ii) Write a chemical equation for the reaction between one molecule of hydrazine, N_2H_4 , and one molecule of water.

..... [1]

[Total: 12]

- 1 The names of the elements of Period 2 of the Periodic Table are shown.

lithium beryllium boron carbon nitrogen oxygen fluorine neon

Answer the following questions about these elements.

Each element may be used once, more than once or not at all.

Identify the element which:

- (a) is a product of photosynthesis

..... [1]

- (b) has an oxide found in clean, dry air

..... [1]

- (c) forms a basic oxide with the formula X_2O

..... [1]

- (d) is a main component of fertilisers used to improve crop growth

..... [1]

- (e) has the highest rate of diffusion at room temperature

..... [1]

- (f) produces a red flame in a flame test

..... [1]

- (g) has only 5 electrons in each of its atoms

..... [1]

- (h) has an oxide responsible for acid rain.

..... [1]

[Total: 8]

- 1 Diamond and graphite are different solid forms of carbon. The carbon atoms in diamond and graphite are arranged in different ways.

(a) State the number of covalent bonds each carbon atom has in diamond.

..... [1]

(b) State the term used to describe the structure of diamond.

..... [1]

(c) Name an oxide that has a similar structure to diamond.

..... [1]

(d) Describe the arrangement of atoms in graphite.

.....

..... [2]

(e) Explain how graphite conducts electricity.

..... [1]

(f) Buckminsterfullerene is a simple molecular form of carbon.

The relative molecular mass of Buckminsterfullerene is 720.

Determine the number of carbon atoms in one molecule of Buckminsterfullerene.

..... [1]

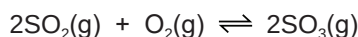
(g) All forms of carbon burn to produce carbon dioxide.

Name the substance used to test for carbon dioxide.

..... [1]

[Total: 8]

- 3 Sulfuric acid is manufactured by an industrial process. Sulfur is obtained from sulfur-containing metal ores.
The sulfur in the metal ore is converted to sulfur dioxide which is then oxidised to sulfur trioxide as shown.



- (a) Name a metal ore which contains sulfur.

..... [1]

- (b) Describe the process which converts metal ores to sulfur dioxide.

..... [1]

- (c) Name the industrial process used to manufacture sulfuric acid.

..... [1]

- (d) The reaction that produces sulfur trioxide is an equilibrium. The forward reaction is exothermic.

- (i) State the temperature and pressure used to make sulfur trioxide.

temperature = °C

pressure = atm
[2]

- (ii) Name the catalyst used.

..... [1]

- (iii) Describe **two** features of an equilibrium.

1

2
[2]

- (iv) State the effect, if any, on the position of equilibrium when the following changes are made.

Explain your answers.

temperature is increased

.....

pressure is increased

.....
[4]

- (v) Explain, in terms of particles, what happens to the rate of reaction when the temperature is increased.

.....
.....
.....
.....
..... [3]

- (e) Name the compound formed when sulfuric acid reacts with ammonia.

..... [1]

[Total: 16]

- 1 Atoms and ions are made from small particles called electrons, neutrons and protons.

(a) Complete the table.

particle	relative charge	relative mass
electron	-1	$\frac{1}{1840}$
neutron		
proton		

[2]

(b) Information about atoms and ions, **A**, **B** and **C**, is shown in the table.

Complete the table.

atom or ion	number of electrons	number of neutrons	number of protons	symbol
A	18		20	${}^{42}_{20}\text{Ca}^{2+}$
B		18		${}^{35}_{17}\text{Cl}$
C	18	16	16	

[6]

[Total: 8]

3 Aluminium is extracted from its ore by electrolysis.

(a) Name the ore of aluminium which consists mainly of aluminium oxide.

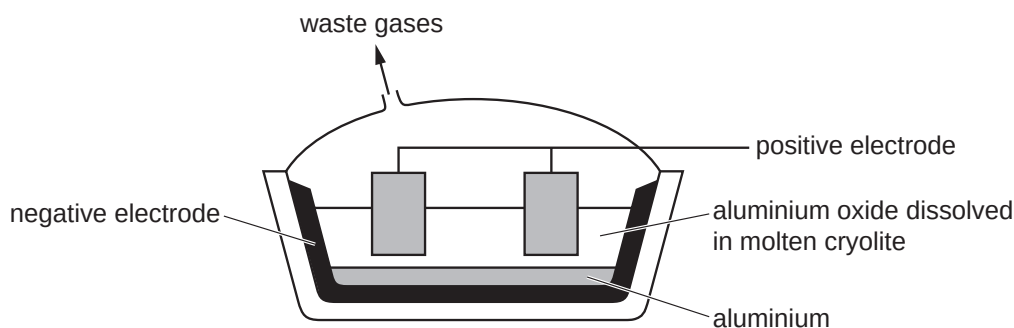
..... [1]

(b) State what is meant by the term *electrolysis*.

.....

..... [2]

(c) Electrolysis is carried out on aluminium oxide dissolved in molten cryolite.



(i) Give **two** reasons why the electrolysis is carried out on aluminium oxide dissolved in molten cryolite instead of electrolysing molten aluminium oxide only.

1

2

[2]

(ii) Write the ionic half-equation for the reaction occurring at the negative electrode.

..... [2]

(iii) The positive electrodes are made of carbon.

Explain why the positive carbon electrodes are replaced regularly.

.....

..... [2]

- (d) Aluminium is more reactive than copper.

When aluminium is added to aqueous copper(II) sulfate, no immediate reaction is seen.

Explain why.

..... [1]

- (e) Aluminium reacts with oxygen to form an amphoteric oxide.

- (i) State what is meant by the term *amphoteric*.

.....
..... [1]

- (ii) The reaction between aluminium oxide and aqueous sodium hydroxide forms a salt containing the negative ion AlO_2^- . The only other product is water.

Write a chemical equation for the reaction between aluminium oxide and aqueous sodium hydroxide.

..... [2]

- (f) Gallium is in the same group as aluminium and forms similar compounds.

Predict the formulae of:

gallium(III) chloride

gallium(III) sulfate. [2]

[Total: 15]

4 This question is about compounds of phosphorus.

- (a) Gaseous phosphorus(V) chloride decomposes into gaseous phosphorus(III) chloride and gaseous chlorine.

When the three gases are present in a closed container the system reaches equilibrium.



- (i) Complete the table using only the words *increases*, *decreases* or *no change*.

	effect on the rate of the forward reaction	effect on the equilibrium yield ($\text{PCl}_3(\text{g})$ and $\text{Cl}_2(\text{g})$)
increasing the temperature		increases
decreasing the pressure		
adding a catalyst		no change

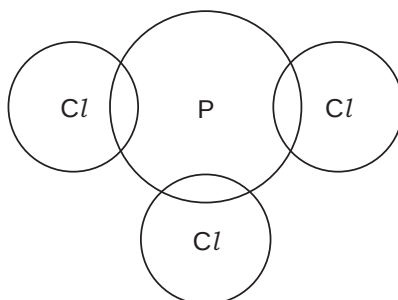
[4]

- (ii) The table shows that when the temperature increases, the equilibrium yields of $\text{PCl}_3(\text{g})$ and $\text{Cl}_2(\text{g})$ increase.

State what conclusion can be made from this.

..... [1]

- (b) Complete the dot-and-cross diagram to show the electron arrangement in a molecule of phosphorus(III) chloride, PCl_3 . Show outer shell electrons only.



[2]

- (c) Phosphorus oxychloride has the formula POCl_3 .

Phosphorus oxychloride is the only product of the reaction between phosphorus(V) chloride, PCl_5 , and phosphorus(V) oxide, P_4O_{10} .

Write a chemical equation for the reaction between phosphorus(V) chloride and phosphorus(V) oxide.

..... [2]

- (d) Compound X has the following composition by mass.

H, 3.66%; P, 37.80%; O, 58.54%

Calculate the empirical formula of compound X.

empirical formula = [2]

- (e) Compound Y has the empirical formula H_3PO_4 and a relative molecular mass of 98.

Deduce the molecular formula of compound Y.

molecular formula = [1]

[Total: 12]

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2 Lithium, sodium and potassium are Group I elements.

(a) Name the type of bonding in these elements.

..... [1]

(b) Sodium reacts with cold water to form hydrogen gas and a solution of a strong alkali.

(i) State the test for hydrogen gas.

test

positive result [1]

(ii) Suggest the pH of a solution of a strong alkali.

pH = [1]

(iii) Name a substance which can be used to confirm the pH of a solution of a strong alkali.

..... [1]

(iv) Write the symbol equation for the reaction between sodium and cold water.

Include state symbols.

..... [3]

(c) Lithium has two naturally occurring types of atoms, ${}^6\text{Li}$ and ${}^7\text{Li}$.

(i) State the name given to atoms of the same element with different nucleon numbers.

..... [1]

(ii) Complete Table 2.1 to show the number of protons, neutrons and electrons in the atom and ion of lithium shown.

Table 2.1

	${}^6\text{Li}$	${}^7\text{Li}^+$
protons		
neutrons		
electrons		

[3]

- (iii) Table 2.2 shows the relative abundance of the two naturally occurring atoms of lithium.

Table 2.2

atom	${}^6\text{Li}$	${}^7\text{Li}$
relative abundance	10%	90%

Calculate the relative atomic mass of lithium to **one** decimal place.

relative atomic mass = [2]

- (d) Potassium oxide, K_2O , is an ionic compound.

Complete Fig. 2.1 to show the electronic configurations of the ions in potassium oxide.
Show the charges on the ions.

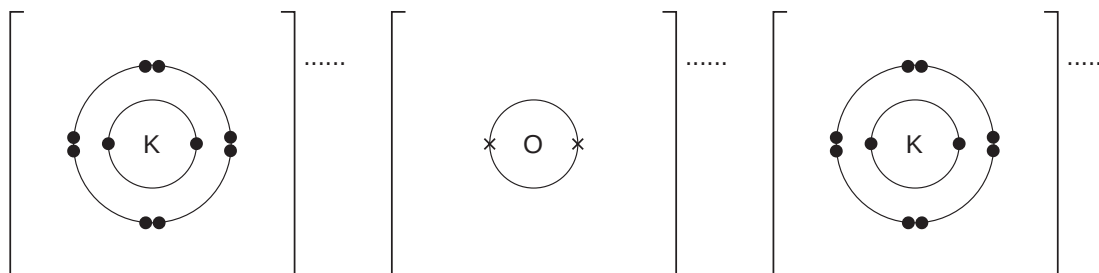


Fig. 2.1

[3]

[Total: 16]

- 2 (a) The symbols of the elements in Period 2 of the Periodic Table are shown.

Li Be B C N O F Ne

Use the symbols of the elements in Period 2 to answer the questions that follow.
Each symbol may be used once, more than once or not at all.

Give the symbol of the element that:

- (i) makes up approximately 78% of clean, dry air [1]
 (ii) contains atoms with only three electrons in the outer shell [1]
 (iii) contains atoms with only nine protons [1]
 (iv) exists as graphite [1]
 (v) is an alkali metal [1]
 (vi) **only** has an oxidation number of zero. [1]

- (b) Boron, B, has two isotopes.

- (i) State the meaning of the term isotopes.

.....
 [2]

- (ii) Table 2.1 shows the relative masses and the percentage abundances of the two isotopes of boron.

Table 2.1

relative mass of isotope	percentage abundance of isotope
10	20
11	80

Calculate the relative atomic mass of boron to **one** decimal place.

relative atomic mass = [2]

[Total: 10]

3 This question is about ionic and covalent compounds.

- (a) (i) Sodium reacts with oxygen to form the ionic compound sodium oxide. The electronic configurations of an atom of sodium and an atom of oxygen are shown in Fig. 3.1.

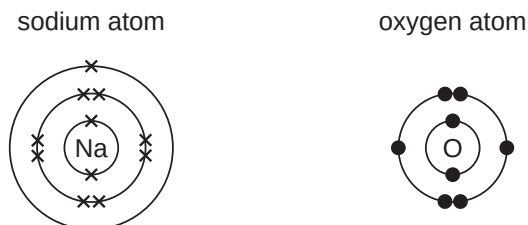


Fig. 3.1

Ions are formed by the transfer of electrons from sodium atoms to oxygen atoms.

Complete the dot-and-cross diagrams in Fig. 3.2 to show the electronic configuration of **one** sodium ion and **one** oxide ion. Show the charges on the ions.

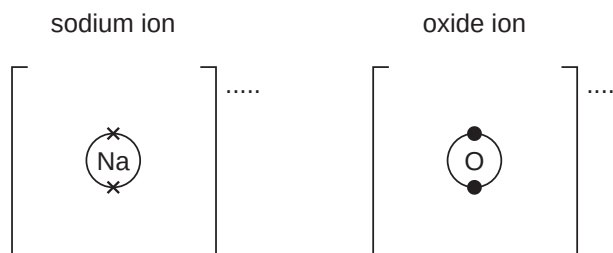


Fig. 3.2

[3]

- (ii) Write the formula of sodium oxide.

..... [1]

- (b) Carbon dioxide, CO_2 , is a covalent compound.

Complete the dot-and-cross diagram in Fig. 3.3 to show the electronic configuration in a molecule of carbon dioxide. Show outer shell electrons only.

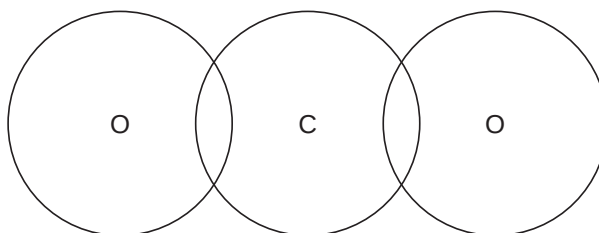


Fig. 3.3

[2]

- (c) The melting points of sodium oxide and carbon dioxide are shown in Table 3.1.

Table 3.1

	melting point/°C
sodium oxide	1275
carbon dioxide	−78

- (i) Explain, in terms of bonding, why sodium oxide has a high melting point.

.....
.....
.....
..... [2]

- (ii) Carbon dioxide has a low melting point.

State the general term for the weak forces that cause carbon dioxide to have a low melting point.

..... [1]

[Total: 9]

2 Fluorine, chlorine and bromine are in Group VII of the Periodic Table.

(a) State the name given to Group VII elements.

..... [1]

(b) Explain why Group VII elements have similar chemical properties.

..... [1]

(c) Complete Table 2.1 to show the colour and state at r.t.p. of some Group VII elements.

Table 2.1

element	colour	state at r.t.p.
fluorine	pale yellow	
chlorine		
bromine		liquid

[3]

(d) Bromine has two naturally occurring isotopes, ^{79}Br and ^{81}Br .

(i) State the term given to the numbers 79 and 81 in these isotopes of bromine.

..... [1]

(ii) Complete Table 2.2 to show the number of protons, neutrons and electrons in the atom and ion of bromine shown.

Table 2.2

	^{79}Br	$^{81}\text{Br}^-$
protons		
neutrons		
electrons		

[3]

- (iii) Table 2.3 shows the relative abundances of the two naturally occurring isotopes of bromine.

Table 2.3

isotope	^{79}Br	^{81}Br
relative abundance	55%	45%

Calculate the relative atomic mass of bromine to **one** decimal place.

relative atomic mass = [2]

- (e) Chlorine displaces bromine from aqueous potassium bromide but does **not** displace fluorine from aqueous sodium fluoride.

- (i) Write the symbol equation for the reaction between chlorine and aqueous potassium bromide.

..... [2]

- (ii) State why chlorine does **not** displace fluorine from aqueous sodium fluoride.

..... [1]

- (f) Aqueous silver nitrate is a colourless solution containing $\text{Ag}^+(\text{aq})$ ions.

- (i) Describe what is seen when aqueous silver nitrate is added to aqueous sodium chloride.

..... [1]

- (ii) Write the ionic equation for the reaction between aqueous silver nitrate and aqueous sodium chloride.

Include state symbols.

..... [3]

[Total: 18]

5 Propane and propene both react with chlorine.

(a) When a molecule of propane, C_3H_8 , reacts with chlorine in the presence of ultraviolet light, one atom of hydrogen is replaced by one atom of chlorine.

(i) State the term given to reactions in which one atom in an alkane is replaced by another atom.

..... [1]

(ii) State the purpose of ultraviolet light in this reaction.

..... [1]

(iii) State the term given to any reaction which requires ultraviolet light.

..... [1]

(iv) Write the symbol equation for the reaction between propane and chlorine.

..... [2]

(b) A molecule of propene, C_3H_6 , is unsaturated and will react with chlorine at room temperature.

(i) State why propene is an unsaturated molecule.

..... [1]

(ii) Give the structural formula of the product of this reaction.

..... [1]

(c) Propene undergoes addition reactions with steam.

There are two possible products, **A** and **B**.

Draw the displayed formula and name each product.

displayed formula of product **A**

name of product **A**

displayed formula of product **B**

name of product **B** [4]

[Total: 11]

- 2 (a) The symbols of the elements in Period 3 of the Periodic Table are shown.

Na Mg Al Si P S Cl Ar

Use the symbols of the elements in Period 3 to answer the questions that follow.
Each symbol may be used once, more than once, or not at all.

Give the symbol of the element that:

- (i) is present in purified bauxite [1]
 (ii) contains atoms with a full outer shell of electrons [1]
 (iii) is used to kill microbes in water treatment [1]
 (iv) forms an amphoteric oxide [1]
 (v) forms an oxide which causes acid rain [1]
 (vi) has an oxidation number of -1 when it forms a compound with hydrogen.
 [1]

- (b) The relative atomic masses of elements can be calculated from the relative masses of isotopes and their percentage abundances.

- (i) Identify the isotope to which all relative masses are compared.
 [1]
 (ii) Table 2.1 shows the relative masses and the percentage abundances of the two isotopes in a sample of magnesium.

Table 2.1

relative mass of isotope	percentage abundance of isotope
24	85
26	15

Calculate the relative atomic mass of magnesium to **one** decimal place.

relative atomic mass = [2]

(c) An ion contains 10 electrons, 13 protons and 14 neutrons.

(i) State the nucleon number of the ion.

..... [1]

(ii) Identify the element that forms this ion.

..... [1]

[Total: 11]

3 Magnesium forms ionic compounds.

- (a) Magnesium reacts with fluorine to form the ionic compound magnesium fluoride. The electronic configurations of an atom of magnesium and an atom of fluorine are shown in Fig. 3.1.

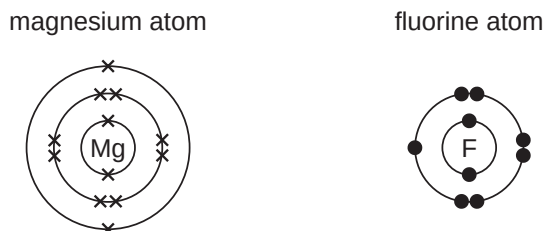


Fig. 3.1

- (i) Ions are formed by the transfer of electrons from magnesium atoms to fluorine atoms.

Complete the dot-and-cross diagrams in Fig. 3.2 to show the electronic configurations of **one** magnesium ion and **one** fluoride ion. Show the charges on the ions.

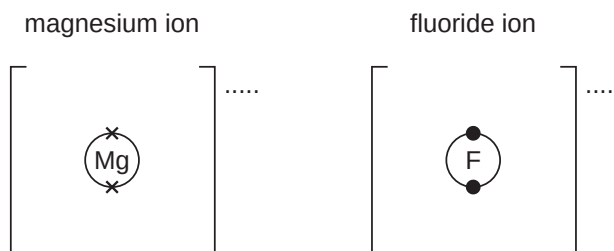


Fig. 3.2

[3]

- (ii) Deduce the formula of magnesium fluoride.

..... [1]

- (iii) When solid magnesium fluoride is dissolved in water it forms a solution that conducts electricity.

State one other change that can be made to solid magnesium fluoride to allow it to conduct electricity.

..... [1]

- (b) Silicon tetrachloride, SiCl_4 , and silicon(IV) oxide, SiO_2 , are covalent compounds.

Complete the dot-and-cross diagram in Fig. 3.3 to show the electronic configuration in a molecule of silicon tetrachloride. Show outer shell electrons only.

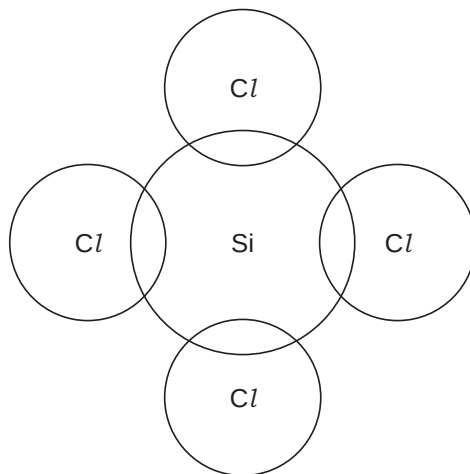


Fig. 3.3

[2]

- (c) The melting points of silicon tetrachloride and silicon(IV) oxide are shown in Table 3.1.

Table 3.1

	melting point/ $^{\circ}\text{C}$
silicon tetrachloride	–69
silicon(IV) oxide	1710

- (i) Silicon tetrachloride has a low melting point because it has weak forces of attraction between particles.

Name the type of particles that are held together by these weak forces of attraction.

..... [1]

- (ii) Explain, in terms of structure and bonding, why silicon(IV) oxide has a high melting point.

.....

 [2]

[Total: 10]

2 Boron and aluminium are Group III elements.

(a) Boron has only two naturally occurring isotopes, ^{10}B and ^{11}B .

Complete Table 2.1 to show the numbers of protons, neutrons and electrons in an atom of ^{11}B .

Table 2.1

number of protons	number of neutrons	number of electrons

[2]

(b) The relative atomic mass of boron to one decimal place is 10.8.

(i) Determine the relative abundance of ^{10}B present in boron. Give your answer as a percentage.

..... % [1]

(ii) Use the relative atomic mass of boron to calculate the number of atoms in 0.540g of boron. Give your answer in standard form.

number of atoms = [2]

(c) Aluminium is extracted from its purified ore as shown in Fig. 2.1.

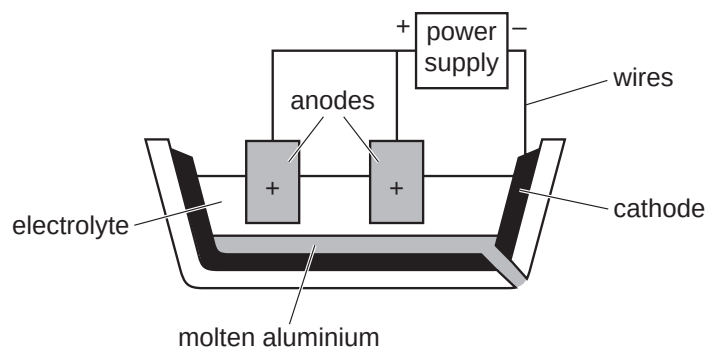


Fig. 2.1

(i) Name the ore of aluminium.

..... [1]

(ii) The electrolyte contains aluminium oxide and one other substance.

Name the other substance and explain why it is used.

name

explanation

..... [2]

(iii) Write the ionic half-equation for the reaction at the cathode.

..... [2]

(iv) Explain why the anodes need frequent replacement.

.....

..... [2]

(d) State **two** physical properties of aluminium that make it suitable for use in overhead electrical cables.

1

2

[2]

(e) Explain the apparent unreactivity of aluminium.

.....
..... [2]

(f) Aluminium reacts with fluorine to form aluminium fluoride, AlF_3 , an ionic compound.

(i) Write the symbol equation for this reaction.

..... [2]

(ii) Complete Fig. 2.2 to show the electronic configuration of one aluminium ion and one fluoride ion.
Show the charges on the ions.

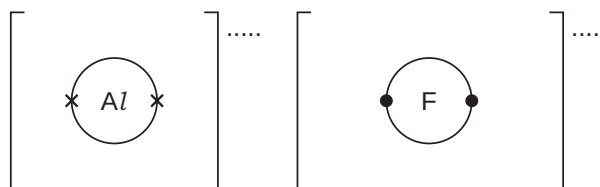


Fig. 2.2

[3]

[Total: 21]

- 1 Table 1.1 gives the electronic configurations of some atoms and ions, **A** to **G**.

Table 1.1

	electronic configuration
A	2,5
B	2,8
C	2,8,2
D	2,8,4
E	2,8,5
F	2,8,6
G	2,8,18,7

Answer the following questions about **A** to **G**.
Each letter may be used once, more than once or not at all.

State which of the atoms or ions, **A** to **G**, could be:

- (a) a noble gas atom

..... [1]

- (b) an atom of an element in Group VI

..... [1]

- (c) an atom with an atomic number of 14

..... [1]

- (d) atoms from the same group

..... and [1]

- (e) a halogen atom

..... [1]

- (f) an atom of an element which is a good conductor of electricity

..... [1]

- (g) a stable ion of a Group V element

..... [1]

- (h) an atom that forms an ion with a 2– charge.

..... [1]

[Total: 8]

2 Cobalt and copper are transition elements.

(a) Copper has two naturally occurring isotopes, ^{63}Cu and ^{65}Cu . Cobalt has only one naturally occurring isotope, ^{59}Co .

(i) Complete Table 2.1 to show the number of protons, neutrons and electrons in the ^{59}Co atom and the $^{65}\text{Cu}^{2+}$ ion.

Table 2.1

	^{59}Co	$^{65}\text{Cu}^{2+}$
protons		
neutrons		
electrons		

[3]

(ii) Table 2.2 shows the relative abundance of the two naturally occurring isotopes of copper.

Table 2.2

isotope	^{63}Cu	^{65}Cu
relative abundance	70%	30%

Calculate the relative atomic mass of copper to **one** decimal place.

relative atomic mass = [2]

(b) One physical property of transition elements such as copper and cobalt is that they are hard. Other metals such as lithium are softer.

State **two** other physical properties of copper and cobalt which are significantly different from lithium.

1

2

[2]

- (c) Both copper and cobalt can form coloured compounds. Some of these compounds contain water of crystallisation.

- (i) Define the term water of crystallisation.

.....
..... [2]

- (ii) State the colour and formula of hydrated cobalt(II) chloride crystals.

colour
formula [2]

- (iii) State the colour change seen when a few drops of water are added to anhydrous copper(II) sulfate.

from to [2]

- (iv) State how this colour change can be reversed.

..... [1]

[Total: 14]

- 2 Table 2.1 gives information about particles **A**, **B**, **C**, **D**, **E** and **F**.

Table 2.1

particle	number of electrons	number of neutrons	number of protons
A	5	6	5
B	10	11	10
C	10	14	13
D	18	17	16
E	18	17	17
F	15	16	15

- (a) Give the letters of **all** the particles which are:

(i) atoms

..... [1]

(ii) ions with a charge of 2–

..... [1]

(iii) cations.

..... [1]

- (b) State the atomic number of **A**.

..... [1]

- (c) Determine the number of nucleons in **D**.

..... [1]

- (d) State the electronic configuration of **D**.

..... [1]

- (e) State the group number of **F**.

..... [1]

- (f) State the period number of **B**.

..... [1]

[Total: 8]

2 Complete Table 2.1.

Table 2.1

atom or ion	number of protons	number of electrons	number of neutrons
$^{37}_{17}\text{Cl}$		17	
$^{63}_{29}\text{Cu}^+$			34
	16	18	17

[5]

3 This question is about the elements sodium and fluorine and the compound sodium fluoride.

(a) Sodium reacts with fluorine to form sodium fluoride.

Write a symbol equation for this reaction.

..... [2]

(b) Some properties of sodium, fluorine and sodium fluoride are shown in Table 3.1.

Table 3.1

	melting point /°C	boiling point /°C	conduction of electricity when solid	conduction of electricity in aqueous solution
sodium	98	883	good	
fluorine	–220	–188	poor	
sodium fluoride	993	1695	poor	good

(i) Explain why sodium conducts electricity when it is a solid.

..... [1]

(ii) Complete the dot-and-cross diagram in Fig. 3.1 of a molecule of fluorine.

Show outer shell electrons only.

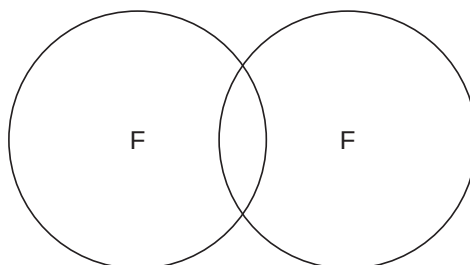


Fig. 3.1

[2]

(iii) Deduce the physical state of fluorine at –200 °C. Use the data in Table 3.1 to explain your answer.

physical state

explanation

.....

[2]

- (iv) Explain in terms of structure and bonding why sodium fluoride has a much higher melting point than fluorine.

.....

.....

.....

.....

..... [3]

- (c) Dilute aqueous sodium fluoride undergoes electrolysis.

Hydrogen is produced at the cathode.

- (i) State what is meant by the term electrolysis.

.....

.....

..... [2]

- (ii) Write an ionic half-equation for the production of hydrogen at the cathode.

..... [2]

[Total: 14]

5 Sulfur is a Group VI element.

(a) A sample of sulfur contains two isotopes, ^{32}S and ^{34}S .

(i) Complete Table 5.1 to show the number of protons and neutrons in one atom of each isotope of sulfur.

Table 5.1

	^{32}S	^{34}S
protons		
neutrons		

[2]

(ii) State why these isotopes have identical chemical properties.

[1]

(iii) State the mass of 6.02×10^{23} atoms of ^{34}S . Include units in your answer.

[1]

(iv) State the name of the amount of substance which contains 6.02×10^{23} atoms.

[1]

(v) Table 5.2 shows the relative abundance of these isotopes of sulfur in the sample.

Table 5.2

atom	^{32}S	^{34}S
relative abundance	95%	5%

Calculate the relative atomic mass of sulfur in this sample to **one** decimal place.

relative atomic mass = [2]

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11

(b) Sulfur reacts with magnesium to form magnesium sulfide, MgS , an ionic compound.

(i) Complete the dot-and-cross diagram in Fig. 5.1 of the ions in magnesium sulfide.

Give the charges on the ions.

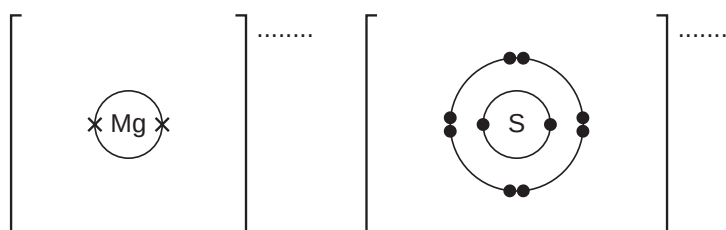


Fig. 5.1

[3]

(ii) State why MgS has a high melting point.

..... [1]

(iii) State why molten MgS conducts electricity.

..... [1]

(c) An acid containing sulfur reacts with sodium hydroxide, NaOH , to form a salt and water. The salt has the formula Na_2SO_3 .

(i) Deduce the formula of this acid.

..... [1]

(ii) Deduce the formula of the anion in Na_2SO_3 .

..... [1]

(d) Na_2SO_3 is oxidised by acidified aqueous potassium manganate(VII).

(i) State what VII refers to in the name potassium manganate(VII).

..... [1]

(ii) State the colour change when this reaction happens.

from to [2]

[Total: 17]

2 Complete Table 2.1.

Table 2.1

atom or ion	number of protons	number of electrons	number of neutrons
${}^{63}_{29}\text{Cu}$	29		
${}^{37}_{17}\text{Cl}^{-}$			20
	30	28	34

[5]

3 This question is about elements and compounds.

(a) Some properties of graphite, oxygen and carbon monoxide are shown in Table 3.1.

Table 3.1

	melting point /°C	boiling point /°C	conduction of electricity when solid
graphite	3652	4827	good
oxygen	−218	−183	poor
carbon monoxide	−199	−191	poor

(i) Explain why graphite conducts electricity when solid.

.....
..... [1]

(ii) Complete the dot-and-cross diagram in Fig. 3.1 of a molecule of oxygen.

Show outer shell electrons only.

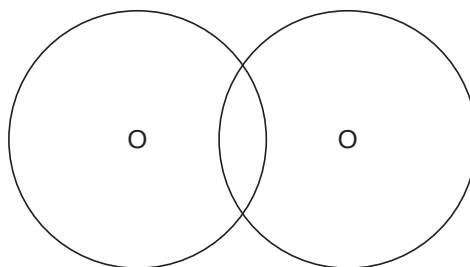


Fig. 3.1

[2]

(iii) Deduce the physical state of carbon monoxide at −195 °C. Use the data in Table 3.1 to explain your answer.

physical state

explanation

..... [2]

- (iv) Explain in terms of structure and bonding why graphite has a much higher melting point than carbon monoxide.

.....
.....
.....
..... [3]

- (b) Potassium reacts with chlorine to form potassium chloride.

Write a symbol equation for this reaction.

..... [2]

- (c) A dilute aqueous solution of potassium chloride undergoes electrolysis.

Oxygen is produced at the anode.

- (i) State what is meant by the term electrolysis.

.....
.....
..... [2]

- (ii) Write an ionic half-equation for the production of oxygen at the anode.

..... [2]

[Total: 14]

2 This question is about atomic structure and the Periodic Table.

(a) Define the term nucleon number.

..... [1]

(b) State the connection between the number of occupied electron shells in an atom and the period number of that element.

..... [1]

(c) Write the electronic configuration of the following atom and ion.

$^{28}_{14}\text{Si}$

$^{37}_{17}\text{Cl}^-$

[2]

(d) Complete Table 2.1.

Table 2.1

atom or ion	number of protons	number of neutrons	number of electrons
$^{23}_{11}\text{Na}$	11		
$^{19}_9\text{F}^-$	9	10	
	31	38	28

[5]



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(e) A sample of thallium, Tl, contains two isotopes, ^{203}Tl and ^{205}Tl .

(i) Define the term isotopes.

.....
..... [2]

(ii) The relative abundance of ^{203}Tl : ^{205}Tl is in the ratio 3:7.

Calculate the relative atomic mass of thallium in the sample to **one** decimal place.

relative atomic mass = [2]

(iii) Suggest why these two isotopes have identical chemical properties.

..... [1]

[Total: 14]



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3 This question is about compounds of tin.

(a) Tin(IV) oxide has the formula SnO_2 .

The relative formula mass, M_r , of SnO_2 is 151.

Calculate the percentage by mass of tin in SnO_2 .

percentage by mass of tin in SnO_2 =% [1]

(b) SnO_2 is an amphoteric oxide.

SnO_2 reacts with aqueous sodium hydroxide, NaOH , to form a sodium salt and water only.
The sodium salt contains a negative ion with the formula SnO_3^{2-} .

(i) State the meaning of the term amphoteric.

.....
..... [1]

(ii) Write the symbol equation for the reaction between SnO_2 and NaOH .

..... [2]

(c) Tin is a metal that forms both covalent and ionic compounds.

Suggest why this is unusual for a metal.

..... [1]





(d) (i) Tin(IV) chloride, SnCl_4 , is covalently bonded.

A tin atom has four electrons in its outer shell.

Complete the dot-and-cross diagram in Fig. 3.1 for a molecule of SnCl_4 .
Show the outer shell electrons only.

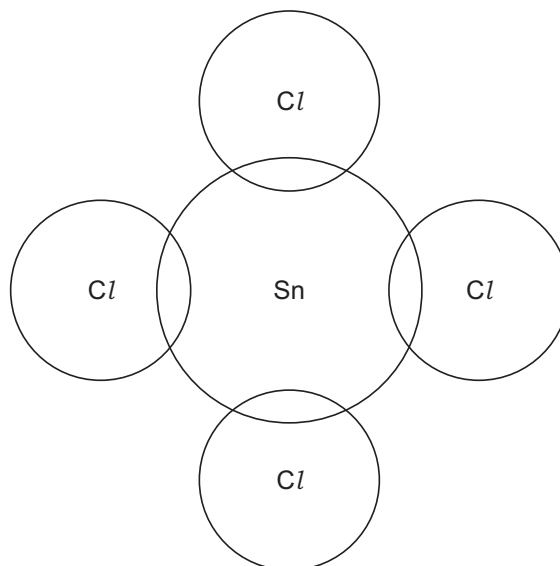


Fig. 3.1

[2]

(ii) Tin(II) oxide, SnO , is ionically bonded.

The melting points of SnCl_4 and SnO are shown in Table 3.1.

Table 3.1

	melting point/ $^{\circ}\text{C}$
SnCl_4	-33
SnO	1080

Explain, in terms of structure and bonding, why SnCl_4 has a much lower melting point than SnO .

.....

.....

.....

.....

..... [3]





(e) Part of the reactivity series is shown.

magnesium	most reactive
aluminium	↑
tin	
copper	least reactive

- (i) When aluminium foil is added to aqueous tin(II) sulfate, a reaction does **not** occur even though aluminium is above tin in the reactivity series.

Explain why a reaction does **not** occur.

.....
 [1]

- (ii) An aqueous solution of tin(II) sulfate contains Sn^{2+} ions.

Two experiments are carried out.

Experiment 1 Copper is added to aqueous tin(II) sulfate.

Experiment 2 Magnesium is added to aqueous tin(II) sulfate.

Write an ionic equation for any reaction that occurs in each experiment.
 If no reaction occurs, write 'no reaction'.

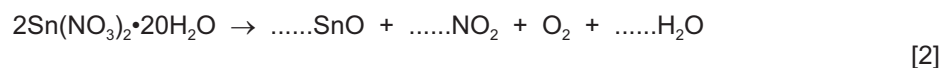
Experiment 1
 Experiment 2 [2]

- (f) Hydrated tin(II) nitrate, $\text{Sn}(\text{NO}_3)_2 \cdot 20\text{H}_2\text{O}$, decomposes when it is heated.

- (i) State what is meant by the term hydrated.

.....
 [1]

- (ii) Complete the equation for the decomposition of $\text{Sn}(\text{NO}_3)_2 \cdot 20\text{H}_2\text{O}$.



[Total: 16]



1 Using numbers only, state the:

- (a) percentage of oxygen in clean, dry air [1]
- (b) typical operating temperature, in °C, used in the Haber process [1]
- (c) number of atoms in a diatomic molecule [1]
- (d) maximum number of electrons in the second electron shell of an atom [1]
- (e) number of hydrogen atoms in an alkane with 7 carbon atoms [1]
- (f) number of particles in **one** mole, in standard form. [1]

[Total: 6]

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2 This question is about ionic compounds.

(a) State what is meant by the term ionic bond.

.....
 [2]

(b) Potassium sulfide, K_2S , is an ionic compound.

Complete the dot-and-cross diagram in Fig. 2.1 of the ions in potassium sulfide.

Show the charges on the ions.

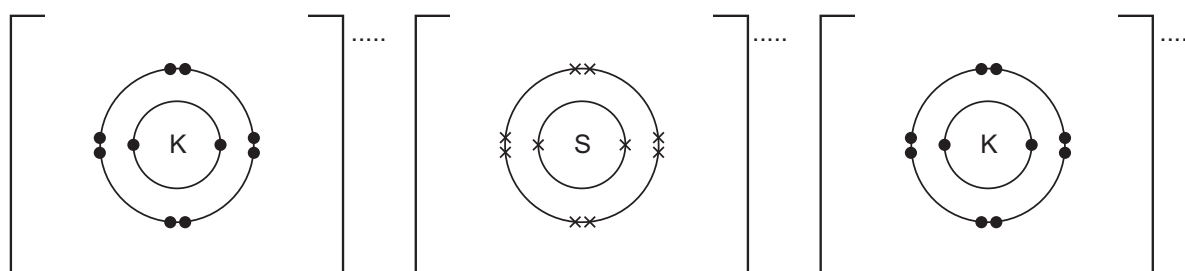


Fig. 2.1

[3]



(c) Ionic compounds form giant ionic lattices.

(i) Fig. 2.2 shows part of the giant ionic lattice structure of sodium chloride.

Complete the diagram in Fig. 2.2 to show the ions present. Use '+' for sodium ions and '-' for chloride ions. One chloride ion has been completed for you.

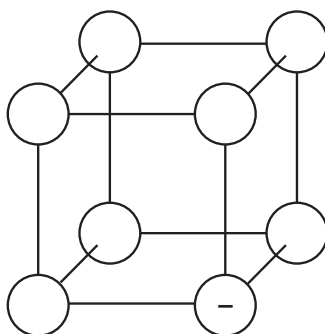


Fig. 2.2

[2]

(ii) State the name given to any positive ion.

..... [1]

(d) Ionic compounds can be decomposed by the passage of an electric current using inert electrodes.

(i) State the name of this process.

..... [1]

(ii) Write the ionic half-equation for the reaction which takes place at the anode when **molten** potassium bromide, KBr, is decomposed by the passage of an electric current.

..... [2]

(iii) Name the products and state the observations at the negative and positive electrodes when **dilute aqueous** potassium bromide, KBr, is decomposed by the passage of an electric current.

product at the negative electrode

.....

observations at the negative electrode

.....

products at the positive electrode

..... and

observations at the positive electrode

.....

[5]



- 5 Manganese is the element with atomic number 25 in the Periodic Table.
Calcium is the element with atomic number 20 in the Periodic Table.

- (a) Complete Table 5.1 to show the number of protons, neutrons and electrons in the ^{55}Mn atom and the $^{42}\text{Ca}^{2+}$ ion.

Table 5.1

	^{55}Mn	$^{42}\text{Ca}^{2+}$
protons		
neutrons		
electrons		

[3]

- (b) Manganese forms several oxides. The formulae of some of these oxides are shown.



- (i) Suggest why manganese is expected to form coloured oxides.
..... [1]
- (ii) State which other property of manganese is shown by the formation of several oxides.
..... [1]
- (iii) State the formula of manganese(II) oxide.
..... [1]
- (c) Mn_3O_4 is found in an ore of manganese. Manganese metal can be extracted from Mn_3O_4 using aluminium as the reducing agent.
- (i) Define the term reducing agent.
.....
..... [2]
- (ii) Complete the symbol equation by inserting the formula of the missing product and balancing the equation.



[2]



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(d) MnO_2 reacts with dilute hydrochloric acid as shown in the equation.



- (i) Calculate the volume of chlorine gas formed, in cm^3 , at r.t.p. when excess MnO_2 reacts with 50.0 cm^3 of $0.200\text{ mol/dm}^3\text{ HCl}$.

Use the following steps.

- Calculate the number of moles of HCl used.

..... mol

- Determine the number of moles of Cl_2 formed.

..... mol

- Calculate the volume of Cl_2 formed.

..... cm^3
[3]

- (ii) Describe a test for chlorine gas.

test

observations
[1]

- (iii) Explain, in terms of collision theory, why decreasing the temperature decreases the rate of this reaction.

.....
.....
.....
.....
.....
.....
..... [3]

[Total: 17]

[Turn over]



2 Atoms are made of electrons, neutrons and protons.

(a) Complete Table 2.1.

Table 2.1

particle	relative charge	relative mass
electron		$\frac{1}{1840}$
neutron	0	
proton		

[2]

(b) Atoms of the same element are known as isotopes.

${}^{39}_{19}\text{K}$ and ${}^{41}_{19}\text{K}$ are isotopes of potassium.

(i) Complete Table 2.2 to show the number of electrons, neutrons and protons in one atom or ion of these isotopes.

Table 2.2

isotope	electrons	neutrons	protons
${}^{39}_{19}\text{K}$			
${}^{41}_{19}\text{K}^+$			

[3]

(ii) Table 2.3 shows the relative masses and the percentage abundances of the two isotopes in a sample of potassium.

Table 2.3

relative mass of isotope	percentage abundance of isotope
39	90
41	10

Calculate the relative atomic mass of this sample of potassium to **one** decimal place.

relative atomic mass = [2]



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(iii) An isotope of aluminium has a nucleon number of 27.

Aluminium has a relative atomic mass of 27.

State what conclusion can be made from this information.

.....
..... [1]

(c) A calcium atom has the electronic configuration of 2,8,8,2.

Give the formula of one atom and one negative ion that has the same electronic configuration as Ca^{2+} .

- atom
- negative ion [2]

[Total: 10]



2 This question is about covalent compounds.

(a) State what is meant by a covalent bond.

.....
..... [2]

(b) Chlorine(I) oxide, Cl_2O , is a simple molecule with covalent bonds.

(i) State what is meant by (I) in the name chlorine(I) oxide.

..... [2]

(ii) Complete the dot-and-cross diagram in Fig. 2.1 of a molecule of chlorine(I) oxide.

Show outer electrons only.

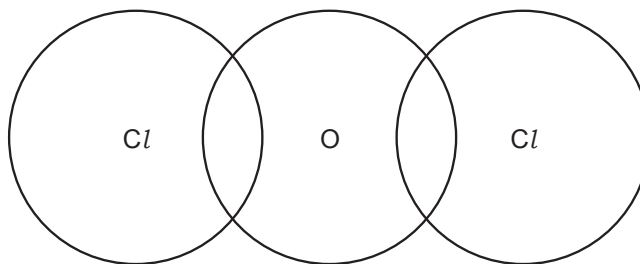


Fig. 2.1

[3]

(iii) Explain, in terms of structure and bonding, why Cl_2O boils at a low temperature and does **not** thermally decompose into its constituent elements, Cl_2 and O_2 .

.....
.....
.....
..... [3]

(iv) Give **two** reasons why liquid Cl_2O is a poor conductor of electricity.

1
2 [2]

(c) Carbon and silicon(IV) oxide both exist as giant covalent structures.

(i) Name a giant covalent structure of carbon which conducts electricity.

..... [1]

(ii) Identify the particles responsible for the conduction of electricity in this covalent structure of carbon.

..... [1]

(iii) Silicon(IV) oxide contains silicon atoms, Si, and oxygen atoms, O.

Fig. 2.2 shows part of the giant covalent structure of silicon(IV) oxide.

Complete the diagram in Fig. 2.2 by adding the symbol for each of the 9 atoms shown.

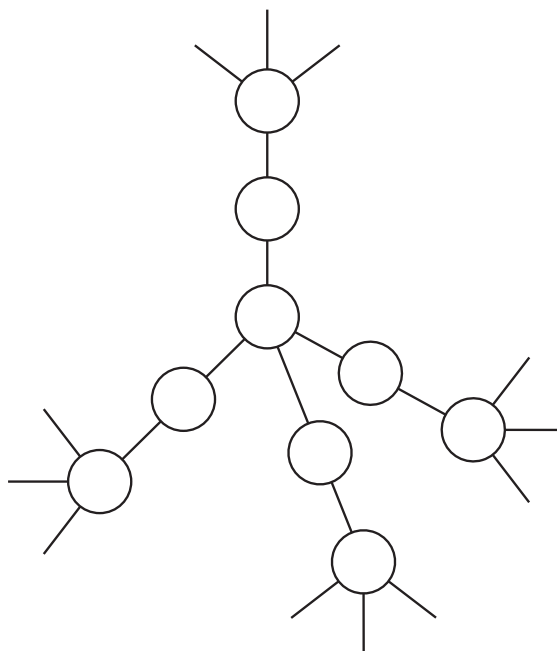


Fig. 2.2

[2]

[Total: 16]

5 Alkali metals are reactive elements.

(a) State the group number of the alkali metals.

..... [1]

(b) Identify the alkali metal which:

(i) has the highest melting point

..... [1]

(ii) has the highest density

..... [1]

(iii) has the lowest reactivity

..... [1]

(iv) burns with a lilac flame

..... [1]

(v) is found in fertilisers to improve plant growth.

..... [1]

(c) Sodium sulfide, Na_2S , is an ionic compound.

Complete Fig. 5.1 to show the electronic configurations of the ions in sodium sulfide.

Show the charges on the ions.

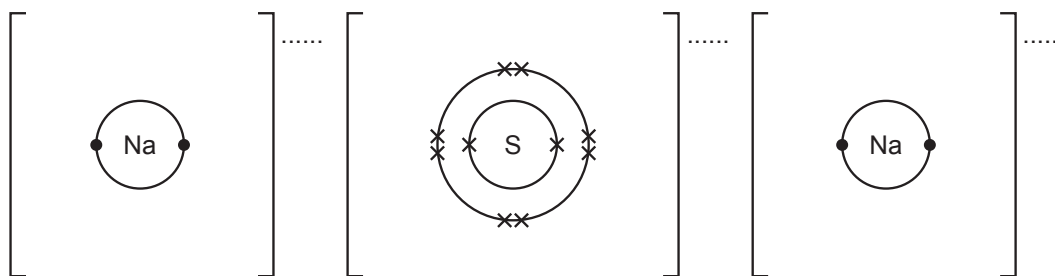


Fig. 5.1

[3]

(d) Rubidium has only two naturally occurring atoms, ^{85}Rb and ^{87}Rb .

(i) State the term given to these naturally occurring atoms of rubidium.

..... [1]

(ii) Complete Table 5.1 to show the number of protons, neutrons and electrons in the atom and ion of rubidium shown.

Table 5.1

	^{85}Rb	$^{87}\text{Rb}^+$
protons		
neutrons		
electrons		

[3]

(iii) The relative atomic mass of rubidium to one decimal place is 85.5.

Determine the relative abundance of ^{85}Rb in rubidium. Express your answer as a percentage.

.....% [1]

[Total: 14]

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- 1 A list of substances is shown.

aluminium oxide	calcium oxide	chlorine	ethanol	graphite
nitrogen	oxygen	propane	propene	silicon(IV) oxide

Answer the questions using the list of substances.

Each substance may be used once, more than once, or not at all.

State which of the substances:

- (a) is a compound with a giant covalent structure
..... [1]
- (b) is an unsaturated hydrocarbon
..... [1]
- (c) is an amphoteric oxide
..... [1]
- (d) is a good conductor of electricity when solid
..... [1]
- (e) contains simple molecules with 9 atoms
..... [1]
- (f) react together to form slag in the blast furnace
..... and [2]
- (g) belongs to a homologous series whose general formula is C_nH_{2n+2}
..... [1]
- (h) is manufactured by the catalytic addition of steam to ethene
..... [1]
- (i) is a gas that is approximately 78% of clean, dry air.
..... [1]

[Total: 10]



2 Atoms are made of electrons, neutrons and protons.

(a) State which of these particles are found in the nucleus of an atom.

..... [1]

(b) Atoms of the same element are known as isotopes.

$^{32}_{16}\text{S}$ and $^{34}_{16}\text{S}$ are isotopes of sulfur.

(i) Complete Table 2.1 to show the number of electrons, neutrons and protons in one atom or ion of these isotopes.

Table 2.1

isotope	electrons	neutrons	protons
$^{34}_{16}\text{S}$			
$^{32}_{16}\text{S}^{2-}$			

[3]

(ii) Table 2.2 shows the relative masses and the percentage abundances of the two isotopes in a sample of sulfur.

Table 2.2

relative mass of isotope	percentage abundance of isotope
32	95
34	5

Calculate the relative atomic mass of this sample of sulfur to **one** decimal place.

relative atomic mass = [2]

(iii) The relative atomic masses of all elements are compared to one atom of an isotope.

Identify this isotope.

..... [1]



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(c) Ions are atoms or groups of atoms that have gained or lost one or more electrons.

An oxygen atom has the electronic configuration of 2,6.

Give the formula of **one** atom and **one** positive ion that has the same electronic configuration as O^{2-} .

• atom

• positive ion

[2]

[Total: 9]



1 Protons, neutrons and electrons are particles found in atoms.

(a) Complete Table 1.1 to show the relative mass and relative charge of a proton, a neutron and an electron.

Table 1.1

particle	relative mass	relative charge
proton		+1
neutron		
electron	$\frac{1}{1840}$	

[2]

(b) Some elements have many isotopes.

(i) Define the term isotopes.

.....
.....
..... [2]

(ii) Explain why all isotopes of the same element have the same chemical properties.

.....
..... [1]

(c) Complete Table 1.2.

Table 1.2

atom or ion	number of protons	number of neutrons	number of electrons
$^{40}_{18}\text{Ar}$	18		18
$^{32}_{16}\text{S}^{2-}$		16	
	22	28	20

[5]



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- (d) The term mass number is defined as the total number of protons and neutrons in the nucleus of an atom.

State the name of **one other** term which is defined as the total number of protons and neutrons in the nucleus of an atom.

..... [1]

- (e) Calculate the number of atoms in 2.00 g of argon.

Give your answer in standard form.

number of atoms = [2]

[Total: 13]



- 1 Atoms and ions are made from particles called electrons, neutrons and protons.

(a) Complete Table 1.1.

Table 1.1

particle	relative charge	relative mass
electron	-1	$\frac{1}{1840}$
neutron		
proton		

[2]

(b) Table 1.2 shows information about some atoms or ions.

Complete Table 1.2.

Table 1.2

atom or ion	number of electrons	number of neutrons	number of protons
${}^{40}_{20}\text{Ca}^{2+}$		20	20
${}^{19}_9\text{F}$	9		9
	10	9	8

[5]

- (c) (i) The relative atomic mass, A_r , of an element is the average mass of its isotopes compared to one atom of isotope X.

Identify isotope X.

..... [1]

- (ii) A sample of magnesium consists of two isotopes with mass numbers 24 and 26. [A_r : Mg, 24.3]

Calculate the percentage abundance of the isotope of magnesium with mass number 24.

percentage abundance = [1]

[Total: 9]



- 2 Table 2.1 shows the melting points, boiling points and electrical conductivities of seven substances, **A, B, C, D, E, F** and **G**.

Table 2.1

substance	melting point / °C	boiling point / °C	electrical conductivity when solid	electrical conductivity when molten
A	1600	2230	poor	poor
B	–219	–183	poor	poor
C	770	1420	poor	good
D	–39	357	good	good
E	–7	58	poor	poor
F	44	280	poor	poor
G	1085	2562	good	good

- (a) Identify which substance, **A, B, C, D, E, F** or **G**, is:

- (i) a gas at 25 °C

..... [1]

- (ii) a solid consisting of simple molecules at 25 °C

..... [1]

- (iii) a metal which is liquid at 25 °C.

..... [1]

- (b) (i) Identify which substance, **A, B, C, D, E, F** or **G**, has a giant covalent structure.

Use the information in Table 2.1 to give **two** reasons for your choice.

substance

reason 1

reason 2

[3]

- (ii) Identify which substance, **A, B, C, D, E, F** or **G**, is an ionic compound.

Use the information in Table 2.1 to give **one** reason for your choice.

substance

reason

[2]

[Total: 8]

